What our results say about error correction

The functional form we fit,

$$R(v) \;=\; F\left(\sum_i w_i \,|d_i^\top v|^n\right),$$

parameterises a family of aggregation rules indexed by n. Two limits are interpretable in error-correction terms. At $n \to \infty$ the inner sum reduces to a max: $R(v) = F(\max_i w_i^{1/n}|d_i^\top v|)$ (up to rescaling of F), so the response is triggered by whichever single projection first exceeds its threshold. This is the signature of *per-feature threshold* error correction: each candidate direction d_i has its own tolerance, and the model tolerates perturbations until one crosses that tolerance. Geometrically, the predicted iso-plateau is a (possibly stretched) square in the d_i basis. At $n = 2$ the inner sum is the quadratic form $\sum_i w_i(d_i^\top v)^2$, so the response depends on a single scalar aggregate of projections across all directions, with no per-direction threshold. The predicted iso-plateau is a (possibly stretched) circle.

Empirically we find $n \approx 2$, with per-pair fit residuals of order 1% across pairs, models, layers, and response metrics; iso-plateaus are approximately circular after normalisation, not square. This distinguishes the two limits at the level of the measured shape. The observation is inconsistent with per-feature threshold error correction: if each candidate direction carried its own independent tolerance, the iso-plateau would approach a square as the tolerances became sharper, not a circle. Our results therefore argue against the hypothesis that the residual stream is protected by tolerances attached to individual feature directions.

We emphasise what is *not* settled by this observation. The $n = 2$ form is itself consistent with error-correcting schemes that threshold on an aggregate quantity (for example, a single global threshold on $\sum_i w_i(d_i^\top v)^2$). Distinguishing such aggregate schemes from a model that simply has a smooth quadratic sensitivity and no error correction of any kind requires inputs beyond the iso-plateau shape. What we have shown is specifically that individual feature directions do not carry individual thresholds.